

Homework 1: Fast-Food Nutrition

Descriptive statistics and data manipulation

Taek Lim

CS 6330: Introduction to Data Science

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Data. Fast foods - Nutritional information.xlsx (126 menu items).

Tooling. Python 3 / pandas / matplotlib; analysis code written by hand, all statistics computed programmatically, none typed by hand; figure code written with Claude Code; ideation manual throughout, including which figures to draw and what each one should show; write-up drafted with Claude Code.

Question 1: Understand the dataset

(15 pts)

Shape and coverage

Measure	Count	Values
Observations	126	menu items (rows)
Variables	8	columns (see Table ??)
Unique restaurants	12	McDonald's, Burger King, Wendy's, Chick-fil-A, Jack in the Box, Sonic, Dairy Queen, Carl's Jr., Hardee's, White Castle, Whataburger, In-N-Out Burger
Unique food types	6	Burger, Milkshake, Breaded Chicken Sandwich, Grilled Chicken Sandwich, Chicken Nuggets, French Fries
Missing values	0	none in any of the 8 columns

Table 1: Shape and coverage of the dataset.

Data type of each variable

Variable	Data type	Attribute type
restaurant	str	Categorical, nominal
item	str	Categorical, nominal
type	str	Categorical, nominal
serving_size	int64	Numeric, ratio
calories	int64	Numeric, ratio
fat	float64	Numeric, ratio
sodium	int64	Numeric, ratio
sugars	float64	Numeric, ratio

Table 2: Variable types. The three categorical variables are nominal: their values can be tested only for equality, and reordering the restaurants or food types would change nothing. The five numeric variables are ratio-scaled, because zero marks the absence of the quantity, so both differences and ratios are meaningful: an 800 kcal item really does carry twice the energy of a 400 kcal one. None of them is interval-scaled, which would require an arbitrary zero. Integer storage does not make a variable discrete either: serving size, calories and sodium are continuous measurements recorded to the nearest unit.

Descriptive statistics

	serving_size (g)	calories (kcal)	sodium (mg)	sugars (g)
count	126	126	126	126
mean	224.27	532.49	973.75	13.29
std	108.03	250.84	523.41	21.02
min	44	130	50	0
25%	126.50	330	569.25	3
median	217.50	515	930	7
75%	315.25	670	1285.25	11
max	467	1240	2460	93
IQR	188.75	340	716.00	8
skew	0.24	0.65	0.42	2.71

Table 3: Descriptive statistics for the four quantitative variables. The skewness row decides which centre to trust: mean and standard deviation are appropriate where $|\text{skew}| < 1$; for **sugars** the mean (13.29) sits *above* the 75th percentile (11), so median and IQR are the honest summary.

Mode

Variable	Mode(s)	Occurrences	Share of 126
serving_size	113 g	4	3.2%
calories	380, 560, 580, 750 kcal (tied)	4 each	3.2% each
sodium	1220 mg	4	3.2%
sugars	0 g	17	13.5%

Table 4: Modal values, with the number of items taking each.

Distribution of each nutrition variable vs. a fitted normal curve (n = 126)

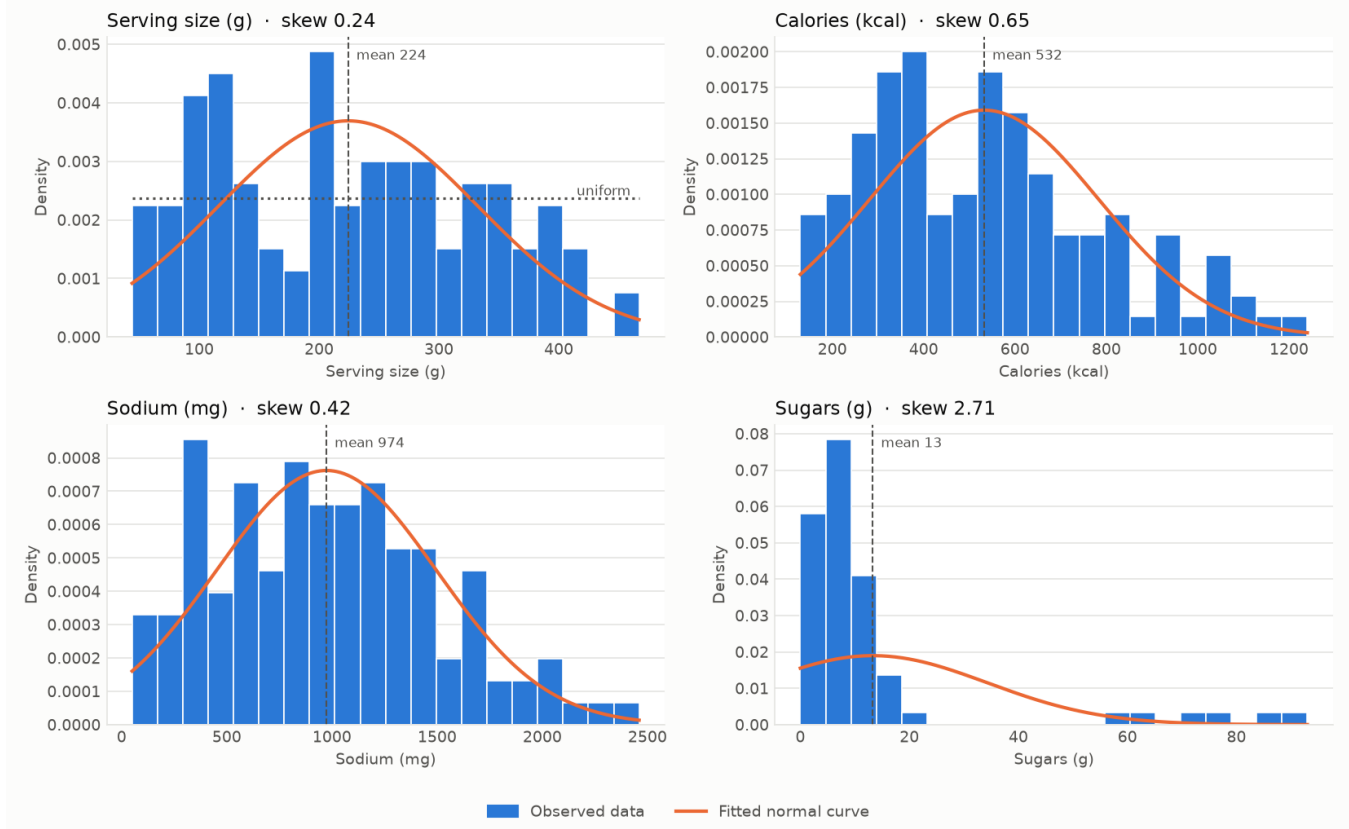


Figure 1: Histograms of the four quantitative variables with a normal curve fitted from the sample mean and standard deviation. `serving_size` is near-symmetric but flat rather than bell-shaped; `calories` and `sodium` are single-peaked with a mild right tail, so a normal approximation is workable; `sugars` is nowhere near normal, since the fitted curve places mass at 30 to 50 g where the data has none.

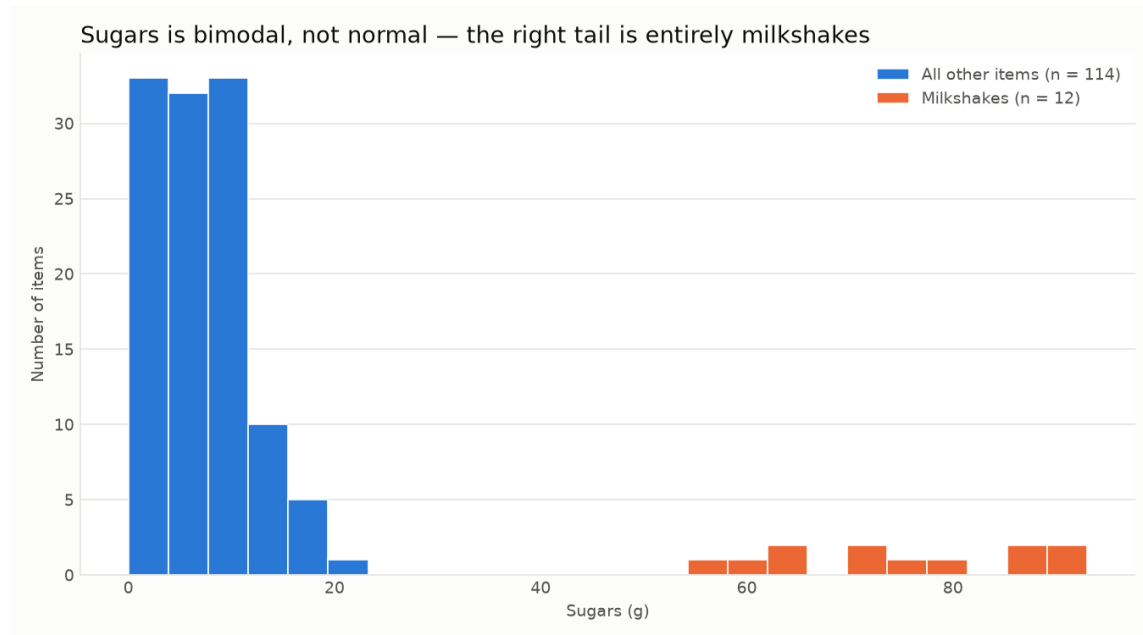


Figure 2: Why `sugars` is skewed. Splitting by food type shows the 2.71 skew is not a long tail but *two separate populations*: 114 items at 0 to 20 g and 12 milkshakes at 56 to 93 g, with no observation at all between 20 and 56 g.

What I learned

About the data:

- The dataset is complete. It holds 126 records across 12 restaurants and 6 food types, with no missing values. A `sugars` reading of 0 g is a real measurement, not a gap.
- The spread is wide. In every quantitative variable the standard deviation is at least half the mean (48%, 47%, 54% and 158% of the mean), so no variable sits close to its centre.
- `sugars` is skewed. Its mean is 13.29 g but the 75th percentile is only 11 g, so a few large values pull the mean above three quarters of the data. Figure ?? shows the cause: the distribution is bimodal.

About the tools: I learned how to pin a Python version for a project and work inside a virtual environment, and how to use pandas to describe a dataset. I also learned that a chart is often the faster way to explain a result, since the shape of `sugars` is obvious in a histogram and invisible in a table of means.

Question 2: Find extreme observations

(12 pts)

Five highest-sodium items

Restaurant	Item	Sodium (mg)	Calories	Type
Hardee's	2/3 LB Double Thickburger	2460	1140	Burger
Hardee's	1/2 LB Original Thickburger	2240	1030	Burger
Jack in the Box	Bacon Ultimate Cheeseburger	2190	910	Burger
Whataburger	A.1. Thick & Hearty Burger	2080	990	Burger
Wendy's	3/4 lb. Triple w/ Cheese	1990	1090	Burger

Table 5: Sorted highest to lowest sodium. All five are burgers. Item names are reproduced from the source file with two spelling corrections (**Thinkbuger** → Thickburger) and trademark symbols dropped; all numbers are unaltered.

Minimum- and maximum-calorie items (all eight variables)

	Restaurant	Item	Type	Serving (g)	Calories	Fat (g)	Sodium (mg)	Sugars (g)
Min	Chick-fil-A	Chicken Nuggets (4pc)	Chicken Nuggets	57	130	6.0	545	1.5
Max	Sonic	SuperSonic Bacon Double CheeseBurger w Mayo	Burger	396	1240	87.0	1690	8.0

Table 6: Complete records (all eight variables) for the minimum- and maximum-calorie items.

Why complete records are more informative

Reporting the complete record lets me compare the calorie extremes on calories per gram rather than raw calories: $1240 \text{ kcal} / 396 \text{ g} \approx 3.13 \text{ kcal/g}$ versus $130 \text{ kcal} / 57 \text{ g} \approx 2.28 \text{ kcal/g}$. The maximum item has 9.54 times the calories but only about 1.4 times the calories per gram, so most of the difference comes from portion size (396 g versus 57 g), and the **fat** variable shows that roughly 63% of its 1240 calories comes from fat alone. The five-variable sodium table omits **serving_size**, so those items cannot be compared per gram, which is exactly the information that a complete record preserves.

Question 3: Filter with multiple conditions

(12 pts)

Subset count

Subset condition: `calories < 500 AND sodium < 1000`, applied as a single mask so both conditions must hold for the same item. **Qualifying items: 49** of 126. Taken separately, 60 items have fewer than 500 calories and 68 have less than 1000 mg sodium, so the joint condition is what reduces the subset to 49.

Qualifying items by restaurant

Restaurant	Qualifying items
Jack in the Box	7
White Castle	7
McDonald's	5
Burger King	5
Wendy's	5
Carl's Jr.	4
Chick-fil-A	3
Sonic	3
Dairy Queen	3
Whataburger	3
Hardee's	2
In-N-Out Burger	2

Table 7: Counts sorted highest to lowest.

Ties. Most counts are tied, so all tied restaurants are reported: 7 (Jack in the Box, White Castle), 5 (McDonald's, Burger King, Wendy's), 3 (Chick-fil-A, Sonic, Dairy Queen, Whataburger), 2 (Hardee's, In-N-Out Burger); only Carl's Jr. at 4 stands alone. Every one of the 12 restaurants has at least one qualifying item, so no restaurant is missing from the table.

What each row represents

One restaurant, and the *number of that restaurant's menu items that satisfy both conditions simultaneously*: fewer than 500 calories *and* less than 1000 mg sodium. It is a count of items, not of restaurants, and it is not a rate: a restaurant with a long menu can show a high count without offering a higher *proportion* of light items.

Question 4: Summarize by food type

(15 pts)

Item counts and calorie statistics by food type

One row per food type, giving the number of menu items (`n`) and the mean, standard deviation, minimum, and maximum of `calories`. The SD is the sample standard deviation (`ddof=1`), which is what `pandas` reports by default in `describe()` and `std()`.

Food type	n	Mean	SD	Min	Max
Burger	69	620.20	277.16	140	1240
Milkshake	12	606.67	98.84	430	800
Breaded Chicken Sandwich	11	522.09	117.04	380	750
Grilled Chicken Sandwich	11	408.00	124.02	180	590
French Fries	12	314.08	49.69	230	395
Chicken Nuggets	11	274.55	119.78	130	530

Table 8: Calories by food type, sorted highest to lowest mean. Sample SD (`ddof=1`).

	n	Mean	SD	Min	Max
Overall	126	532.49	250.84	130	1240

Table 9: Calories over all 126 items pooled.

Two observations

Observation 1. Burgers have the highest mean calories (620.20), but they are also the most widely spread type, with a standard deviation of 277.16 and a range from 140 to 1240 calories. Because the spread is this large, knowing only that an item is a burger does not tell me much about its calorie count.

Observation 2. Milkshakes average almost as high as burgers (606.67 vs. 620.20, a gap of 13.53) but with a third of the variability (SD 98.84 vs. 277.16, range 430 to 800). A milkshake is *reliably* calorie-dense, whereas a burger may be light (140) or extreme (1240). The two means are close; the two distributions are not.

Question 5: Summarize by restaurant

(15 pts)

Menu items and mean calories, sodium, and sugars by restaurant

Restaurant	n	Mean calories	Mean sodium (mg)	Mean sugars (g)
Hardee's	12	691.67	1399.17	14.92
Whataburger	12	603.33	1149.17	14.75
Sonic	11	588.18	926.36	10.00
Carl's Jr.	13	585.31	1085.92	12.99
Wendy's	12	549.17	1057.50	14.17
Jack in the Box	13	541.54	1082.31	11.69
Burger King	11	540.00	817.27	14.00
Dairy Queen	11	516.36	1001.82	12.09
In-N-Out Burger	5	505.00	731.00	19.00
McDonald's	11	441.82	766.36	12.45
Chick-fil-A	5	330.00	654.00	18.50
White Castle	10	319.00	568.00	10.60

Table 10: Sorted by mean calories, highest to lowest.

Highest and lowest restaurant on each mean measure

- **Mean calories:** highest = Hardee's (691.67), lowest = White Castle (319.00).
- **Mean sodium:** highest = Hardee's (1399.17), lowest = White Castle (568.00).
- **Mean sugars:** highest = In-N-Out Burger (19.00), lowest = Sonic (10.00).

Why these summaries are not comparisons of equivalent menus

These means describe only the items from each restaurant that are in this dataset, not the whole menu, and the menus are not made up of the same things: Chick-fil-A has no burger here, so its low mean (330.00) mostly shows that the highest-calorie food type is missing from its list, not that its food is lighter. The number of items also differs a lot, from 5 for In-N-Out Burger to 13 for Carl's Jr., so a mean based on only 5 items is much less reliable than one based on 13. To compare restaurants fairly, I would have to compare the same food type across restaurants.

Question 6: Use two grouping variables

(23 pts)

Mean calories for every restaurant $\times$ food-type combination, and the lowest-mean restaurant in each type

Food type	Restaurant	Mean cal.	n
Breaded Chicken Sandwich	In-N-Out Burger	0.00	0
Breaded Chicken Sandwich	White Castle	380.00	1
Breaded Chicken Sandwich	Chick-fil-A	410.00	1
Breaded Chicken Sandwich	Jack in the Box	410.00	1
Breaded Chicken Sandwich	Whataburger	430.00	1
Breaded Chicken Sandwich	Carl's Jr.	493.00	1
Breaded Chicken Sandwich	McDonald's	510.00	1
Breaded Chicken Sandwich	Wendy's	510.00	1
Breaded Chicken Sandwich	Sonic	580.00	1
Breaded Chicken Sandwich	Dairy Queen	610.00	1
Breaded Chicken Sandwich	Burger King	660.00	1
Breaded Chicken Sandwich	Hardee's	750.00	1
Burger	Chick-fil-A	0.00	0
Burger	White Castle	244.00	5
Burger	McDonald's	508.33	6
Burger	In-N-Out Burger	513.33	3
Burger	Dairy Queen	571.67	6
Burger	Jack in the Box	590.00	8
Burger	Burger King	626.67	6
Burger	Wendy's	662.86	7
Burger	Carl's Jr.	686.75	8
Burger	Sonic	690.00	6
Burger	Whataburger	738.57	7
Burger	Hardee's	804.29	7
Chicken Nuggets	In-N-Out Burger	0.00	0
Chicken Nuggets	Chick-fil-A	130.00	1
Chicken Nuggets	Burger King	170.00	1
Chicken Nuggets	Wendy's	180.00	1
Chicken Nuggets	McDonald's	190.00	1
Chicken Nuggets	Carl's Jr.	220.00	1
Chicken Nuggets	Jack in the Box	240.00	1
Chicken Nuggets	Hardee's	260.00	1
Chicken Nuggets	Dairy Queen	350.00	1
Chicken Nuggets	Whataburger	370.00	1
Chicken Nuggets	Sonic	380.00	1
Chicken Nuggets	White Castle	530.00	1
Overall		532.49	126

Food type	Restaurant	Mean cal.	n
French Fries	McDonald's	230.00	1
French Fries	Sonic	250.00	1
French Fries	Chick-fil-A	280.00	1
French Fries	Whataburger	280.00	1
French Fries	Carl's Jr.	304.00	1
French Fries	Wendy's	310.00	1
French Fries	Burger King	320.00	1
French Fries	Jack in the Box	330.00	1
French Fries	White Castle	330.00	1
French Fries	Hardee's	360.00	1
French Fries	Dairy Queen	380.00	1
French Fries	In-N-Out Burger	395.00	1
Grilled Chicken Sandwich	In-N-Out Burger	0.00	0
Grilled Chicken Sandwich	White Castle	180.00	1
Grilled Chicken Sandwich	Chick-fil-A	270.00	1
Grilled Chicken Sandwich	McDonald's	350.00	1
Grilled Chicken Sandwich	Dairy Queen	360.00	1
Grilled Chicken Sandwich	Wendy's	370.00	1
Grilled Chicken Sandwich	Carl's Jr.	398.00	1
Grilled Chicken Sandwich	Burger King	420.00	1
Grilled Chicken Sandwich	Sonic	450.00	1
Grilled Chicken Sandwich	Jack in the Box	540.00	1
Grilled Chicken Sandwich	Whataburger	560.00	1
Grilled Chicken Sandwich	Hardee's	590.00	1
Milkshake	Whataburger	430.00	1
Milkshake	McDonald's	530.00	1
Milkshake	Dairy Queen	550.00	1
Milkshake	White Castle	550.00	1
Milkshake	Chick-fil-A	560.00	1
Milkshake	Wendy's	580.00	1
Milkshake	In-N-Out Burger	590.00	1
Milkshake	Burger King	610.00	1
Milkshake	Sonic	670.00	1
Milkshake	Carl's Jr.	700.00	1
Milkshake	Hardee's	710.00	1
Milkshake	Jack in the Box	800.00	1
Overall		532.49	126

Table 11: Mean calories for the full $6 \times 12 = 72$ restaurant $\times$ food-type grid, sorted by food type and then by mean calories. The four combinations that do not appear in the menu data are shown as 0.00 with $n = 0$. The table continues in the right column.

The four combinations shown as 0:

- In-N-Out Burger, Breaded Chicken Sandwich
- In-N-Out Burger, Grilled Chicken Sandwich
- In-N-Out Burger, Chicken Nuggets
- Chick-fil-A, Burger

Within each food type the minimum of those means is taken, and every restaurant attaining it is returned (see *Tie handling* below).

Lowest mean calories per food type (six rows)

Food type	Restaurant	Mean calories	n items
Breaded Chicken Sandwich	White Castle	380.00	1
Burger	White Castle	244.00	5
Chicken Nuggets	Chick-fil-A	130.00	1
French Fries	McDonald's	230.00	1
Grilled Chicken Sandwich	White Castle	180.00	1
Milkshake	Whataburger	430.00	1

Table 12: Lowest mean calories per food type. Note the n column: five of the six “winners” rest on a single menu item, so only the Burger row (White Castle, $n = 5$) is a stable estimate.

Tie handling

My procedure finds the minimum mean within each food type and returns every restaurant whose mean is equal to that minimum. Three ways of doing this give the same six rows on this dataset:

```
1. g.loc[g.groupby("type").mean_cal.idxmin()]
2. g.sort_values("mean_cal").groupby("type").head(1)
3. g[g.mean_cal == g.groupby("type").mean_cal.transform("min")]
```

The first two keep only one row per food type, so if two restaurants had the same lowest mean, one of them would be dropped without any error message. The third one compares every mean against the group minimum, so it returns all tied restaurants and the table would have more than six rows. No tie occurs in this dataset, so all three happen to be correct here, but on a larger dataset ties do happen, and silently dropping a tied record means losing information and possibly making a decision based on an incomplete answer. That is why I used the third method.

Question 7: Debug an analysis

(8 pts)

Mean calories for every restaurant

Restaurant	Mean calories
Hardee's	691.67
Whataburger	603.33
Sonic	588.18
Carl's Jr.	585.31
Wendy's	549.17
Jack in the Box	541.54
Burger King	540.00
Dairy Queen	516.36
In-N-Out Burger	505.00
McDonald's	441.82
Chick-fil-A	330.00
White Castle	319.00

Table 13: Mean calories by restaurant, highest to lowest. The answer to the question is the first row of this table.

Result of the given code

```
df.groupby('restaurant')['calories'].mean().sort_values().head(1)

restaurant
White Castle 319.00
```

Listing output is rounded to two decimal places throughout, matching the tables.

The R version has the same defect:

```
df |>
  group_by(restaurant) |>
  summarise(mean_calories = mean(calories)) |>
  arrange(mean_calories) |>
  slice_head(n = 1)
```

Both return the *last* row of the table above, not the first one. The code runs without any error message, so the wrong answer is easy to miss.

Corrected code

```
df.groupby('restaurant')['calories'].mean().sort_values(ascending=False).head(1)

df |>
  group_by(restaurant) |>
  summarise(mean_calories = mean(calories)) |>
  arrange(desc(mean_calories)) |>
  slice_head(n = 1)
```

Result of the corrected code

```
restaurant
Hardee's 691.67
```

This matches the first row of the table above.

Why the original code was wrong

`sort_values()` sorts in ascending order by default, so `head(1)` took the smallest mean instead of the largest. `arrange()` in R defaults to ascending in the same way, so `slice_head(n = 1)` has the same problem. Adding `ascending=False`, or `desc()` in R, sorts from highest to lowest, and the first row is then the restaurant the question asks for. The grouping and the mean were already correct.

A problem that remains after the fix

`head(1)` returns one row no matter what, so if two restaurants had the same highest mean, one of them would be dropped without any warning. Comparing every mean against the maximum returns all of them instead, which is the same tie rule I used in Question 6:

```
s = df.groupby('restaurant')['calories'].mean()
s[s == s.max()]
```

```
df |>
  group_by(restaurant) |>
  summarise(mean_calories = mean(calories)) |>
  filter(mean_calories == max(mean_calories))
```

No tie occurs in this dataset, so this returns exactly one row, Hardee's at 691.67.